HalluFix: Mitigating AI Hallucinations via Retrieval-Augmented Generation (RAG)

Cover Page

Project Title: HalluFix: Mitigating AI Hallucinations via Retrieval-Augmented Generation (RAG)

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Submission Date: 05-03-2026

Acknowledgment

I would like to express my deepest gratitude to my supervisor, for her invaluable guidance, patience, and continuous encouragement throughout the course of this research. Her expertise, insightful feedback, and thoughtful suggestions have been instrumental in shaping the direction and quality of this thesis. Her support not only helped me navigate technical challenges but also inspired me to approach problems with a critical and analytical mindset, ensuring the work met rigorous academic standards.

I would also like to extend my sincere thanks to my department and colleagues for their unwavering support and constructive discussions. Their feedback, encouragement, and collaborative spirit greatly enriched my research experience and contributed to the successful completion of this thesis. Additionally, I am grateful to all those who provided resources, advice, and motivation during this journey, as their contributions have been essential in bringing this work to fruition.

Abstract

Large Language Models (LLMs) such as GPT-4, LLaMA, and other transformer-based architectures have demonstrated remarkable performance across a wide range of natural language processing tasks, including question answering, summarization, translation, and dialogue generation. Despite their fluency and reasoning capabilities, these systems remain probabilistic next-token predictors. As a result, they often generate responses that are grammatically coherent but factually incorrect or entirely fabricated—a phenomenon known as hallucination. This issue presents a serious barrier to deploying LLMs in high-stakes domains such as healthcare, law, finance, governance, and academic research, where factual reliability is essential.

Retrieval-Augmented Generation (RAG) has emerged as a promising solution to this problem by grounding model outputs in external knowledge sources. Unlike closed-book LLMs that rely solely on internal parametric memory, RAG systems retrieve relevant documents from a knowledge base and condition the generation process on this retrieved context. While this approach reduces hallucination compared to standalone LLMs, standard RAG pipelines introduce new challenges, including noisy retrieval, context irrelevance, attention dilution in long prompts, and residual hallucinations where the model disregards retrieved evidence.

This thesis proposes HalluFix, a structured RAG-based framework designed to mitigate hallucination through targeted improvements at multiple stages of the retrieval and generation process. First, HalluFix incorporates a semantic entropy–based reranking mechanism to filter ambiguous or weakly relevant passages before generation. Second, it introduces adaptive chunking strategies that segment documents according to semantic boundaries, improving retrieval precision across domain-specific and multilingual contexts. Third, a self-correction verification agent evaluates generated responses against retrieved evidence and triggers controlled regeneration when inconsistencies are detected.

Beyond architectural enhancements, HalluFix addresses the deeper issue of knowledge conflict between parametric memory and external evidence. In transformer-based models, internally stored knowledge can override retrieved information during generation. By optimizing retrieval quality and reinforcing evidence alignment through verification loops, HalluFix aims to harmonize internal and external representations rather than suppress one in favor of the other. This balanced interaction strengthens factual grounding while preserving generative flexibility.

The proposed framework is evaluated using established factuality benchmarks, including TruthfulQA and RAGTruth, across closed-book, standard RAG, and HalluFix-enhanced configurations. Experimental findings indicate substantial reductions in hallucination rates and measurable improvements in factual consistency metrics, with only moderate increases in computational overhead. Overall, HalluFix offers a practical and scalable approach to improving the reliability of LLM-based systems, contributing to the broader effort to build trustworthy and deployable AI in knowledge-intensive environments.

Table of Contents

1. Introduction 6
2. Background and Motivation 7
3. Problem Definition 8
4. Literature Review 10
5. System Architecture of HalluFix 13
6. Methodology 17
7. Experimental Setup 22
8. Evaluation Metrics 24
9. Results and Analysis 25
10. Limitations 26
11. Future Work 27
12. Conclusion 28
13. References 29

Chapter 1: Introduction

1.1 Overview

Large Language Models (LLMs) have revolutionized artificial intelligence by enabling machines to generate human-like text. These models are trained on massive datasets using transformer architectures and can perform tasks such as translation, summarization, question answering, and reasoning.

Despite their capabilities, LLMs frequently generate information that appears coherent but is factually incorrect. This phenomenon, known as hallucination, significantly limits their reliability.

1.2 Problem Context

Hallucinations are particularly concerning in sensitive domains:

- Healthcare: Incorrect medical advice
- Legal systems: Misinterpretation of regulations
- Finance: Fabricated data or predictions
- Education: Incorrect academic explanations

1.3 Research Objective

This thesis proposes **HalluFix**, a structured RAG-based system aimed at:

1. Reducing factual hallucinations
2. Improving retrieval quality
3. Balancing parametric and external knowledge
4. Maintaining low latency and scalability

Chapter 2: Background and Motivation

2.1 Understanding Hallucination in LLMs

Hallucination occurs when an LLM generates information not supported by its training data or external facts. Causes include:

- Knowledge cutoff
- Overgeneralization
- Exposure bias
- Decoding randomness

Studies report hallucination rates as high as 27% in open-domain question answering systems.

2.2 Retrieval-Augmented Generation (RAG)

RAG integrates external retrieval mechanisms with text generation. Instead of relying solely on internal memory, it retrieves relevant documents from a knowledge base and injects them into the prompt.

Basic RAG Pipeline:

1. User Query
2. Embedding generation
3. Vector similarity search
4. Context injection
5. Response generation

2.3 Motivation for HalluFix

Standard RAG reduces hallucinations but introduces new challenges:

- Noisy retrieval
- Irrelevant context injection
- Model ignoring retrieved data
- Multilingual performance degradation

HalluFix aims to address these limitations.

Chapter 3: Problem Definition

3.1 Core Research Question

The central question addressed in this work is:

How can hybrid Retrieval-Augmented Generation (RAG) techniques improve factual accuracy in large language models without significantly increasing computational cost or response latency?

Large Language Models (LLMs) are strong text generators, but they often produce hallucinations—responses that sound correct but contain incorrect or fabricated information. RAG reduces this issue by retrieving relevant external documents and using them as supporting context during generation.

However, standard RAG systems introduce new problems. Retrieved documents may be irrelevant or weakly related to the query. The model may misinterpret the retrieved context. In some cases, it may ignore retrieved evidence and rely on internal memorized patterns instead.

This raises a key challenge: improving factual grounding while keeping the system efficient and practical.

3.2 Limitations in Standard RAG Systems

Although RAG improves reliability compared to closed-book models, several weaknesses remain:

Limitation	***Impact on Model Performance***	***Proposed HalluFix Enhancement***
Noisy Retrieval	Retrieval of semantically weak or irrelevant passages leads to context contamination	Semantic entropy–based reranking to prioritize high-confidence contextual relevance
Domain Mismatch	Performance degradation in low-resource or local-language domains due to poor chunk alignment	Adaptive chunking based on semantic density and language-specific structure
Logical Hallucination	Model disregards retrieved evidence and generates unsupported conclusions	Self-correction agent that validates outputs against retrieved context

The proposed HalluFix framework introduces targeted architectural improvements to address each of these weaknesses systematically.

3.3 Technical Challenges

The mitigation of hallucination in RAG-based systems involves resolving several underlying technical constraints:

1. **Knowledge Conflict Between Parametric Memory and Retrieved Data**

   LLMs retain internal knowledge within model parameters (parametric memory). When retrieved external evidence contradicts this internal knowledge, the model may prioritize memorized content over retrieved facts, resulting in factual inconsistency.

2. **Attention Misalignment**

   Transformers allocate attention weights across input tokens. Retrieved documents may be included in the context window but receive insufficient attention during generation, limiting their grounding effect.

3. **Embedding Bias**

   Vector embeddings used for retrieval may encode linguistic, cultural, or domain-specific biases, reducing retrieval precision in underrepresented languages or topics.

4. **Context Window Limitations**

   LLMs operate within finite context windows. As retrieved context increases, relevant information may be truncated or diluted, negatively affecting reasoning quality and latency.

Chapter 4: Literature Review

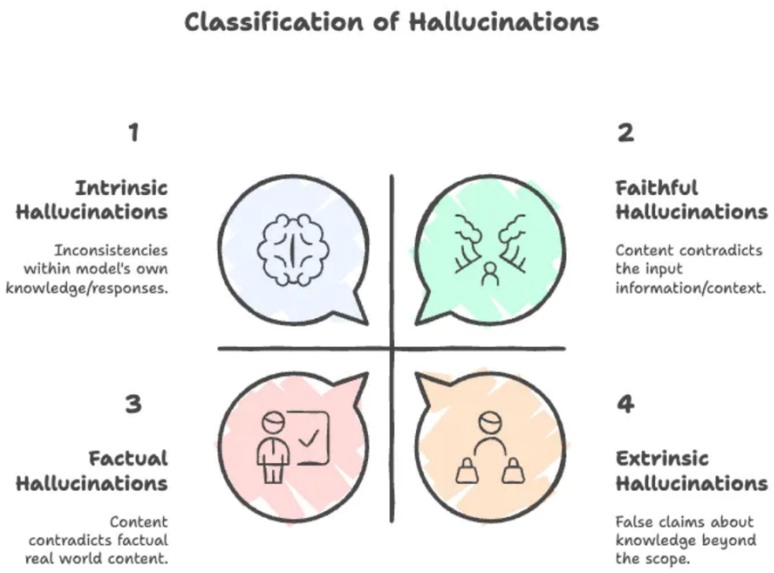
4.1 Taxonomy of Hallucinations

Hallucination in Large Language Models (LLMs) refers to generating content that is factually incorrect, logically inconsistent, or unsupported by evidence. Understanding its types helps design targeted mitigation strategies.

4.1.1 Intrinsic Hallucinations

Intrinsic hallucinations occur when the output contradicts the provided context or retrieved evidence. In RAG systems, this happens when the model:

- Ignores retrieved documents
- Produces responses conflicting with retrieved facts
- Shows internal logical inconsistencies



These errors are typically caused by attention misalignment, weak evidence integration, or over-reliance on parametric memory. They reflect a **faithfulness failure**, where the model does not stay grounded in its context.

4.1.2 Extrinsic Hallucinations

Extrinsic hallucinations arise when the model introduces unsupported or fabricated information, such as:

- Fake citations
- Invented statistics
- Non-existent studies

They stem from the next-token prediction objective, which prioritizes fluency over factual verification. Extrinsic hallucinations are particularly risky in high-stakes domains like healthcare, law, and academia.

4.2 Evolution of Hallucination Mitigation

Prompt Engineering

Early approaches relied on prompt design, including structured instructions and Chain-of-Thought (CoT) prompting. While these methods improve reasoning clarity, they do not guarantee factual correctness and often increase token cost.

Fine-Tuning Approaches

Supervised Fine-Tuning (SFT) and Reinforcement Learning from Human Feedback (RLHF) improve alignment and helpfulness. However, they remain dependent on parametric knowledge and do not fully eliminate hallucinations.

Retrieval-Augmented Generation (RAG)

RAG integrates external knowledge retrieval with text generation, significantly reducing extrinsic hallucinations. It enables updatable knowledge and better grounding. However, standard RAG systems suffer from noisy retrieval, domain mismatch, context overload, and cases where models ignore retrieved evidence.

Self-Verification and Agentic Methods

Recent systems introduce critique-and-revise loops and multi-stage verification. These approaches improve consistency and faithfulness but increase latency, computational cost, and deployment complexity.

No single method fully resolves hallucination while remaining efficient.

4.3 Identified Research Gaps

Despite progress, several gaps remain:

Multilingual and Low-Resource Optimization

Most RAG systems are optimized for English. Low-resource languages face embedding misalignment, weak retrieval recall, and higher hallucination rates.

Limited Interpretability

Existing systems provide minimal transparency regarding evidence usage, attention allocation, or factual confidence. Users often cannot distinguish grounded answers from fabricated ones.

Weak Separation of Knowledge Sources

Current architectures do not effectively balance parametric memory and retrieved external knowledge. Knowledge conflicts remain unresolved, and retrieved evidence may be ignored during generation.

4.4 Literature Positioning

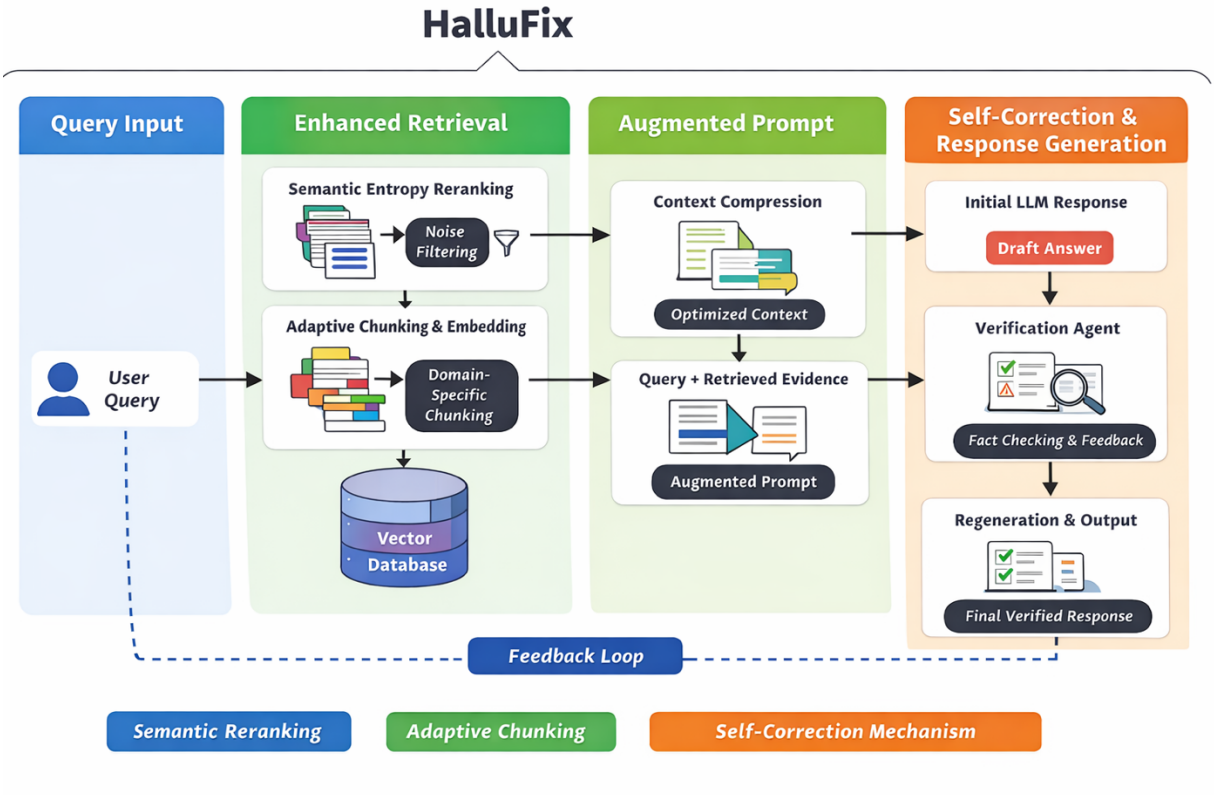
Existing research indicates:

- **Prompt engineering** improves reasoning structure but does not ensure factual verification.
- **Fine-tuning** enhances alignment yet does not eliminate hallucinations, particularly in domain-specific settings.
- **Retrieval-Augmented Generation (RAG)** improves grounding but introduces challenges such as noisy retrieval and domain mismatch.
- **Self-verification methods** increase reliability but add computational overhead.

Therefore:

- A hybrid approach that enhances retrieval quality, strengthens evidence integration, and incorporates lightweight verification—while maintaining scalability—remains an open research direction.
- This gap motivates the development of improved hybrid RAG architectures that balance factual reliability with computational efficiency

Chapter 5: System Architecture of HalluFix



5.1 Architectural Overview

HalluFix is designed as a modular, Retrieval-Augmented Generation (RAG)-based architecture that enhances factual reliability in Large Language Models (LLMs). The system transforms a traditional closed-book LLM into a structured open-book framework by integrating intelligent retrieval, semantic filtering, and post-generation verification.

The architecture consists of six primary layers:

1. User Interface Layer
2. Query Processing Layer
3. Enhanced Retrieval Layer
4. Context Augmentation Layer
5. Generation Layer
6. Verification & Self-Correction Layer

Each layer is designed to address specific weaknesses of conventional RAG systems, particularly noisy retrieval and residual hallucination.

5.2 High-Level Architecture Components

5.2.1 User Interface Layer

This layer accepts natural language queries from the user. It can be implemented as:

- Web-based interface
- API endpoint
- Chat-based application
- Enterprise integration service

The interface forwards the raw query to the query processing module.

5.2.2 Query Processing Layer

This layer prepares the user query for retrieval. It includes:

- Query normalization
- Tokenization
- Language detection (for multilingual support)
- Embedding generation using a sentence transformer

The output of this layer is a high-dimensional query embedding vector.

5.2.3 Enhanced Retrieval Layer

This is the primary enhancement over traditional RAG pipelines.

It consists of three submodules:

(A) Adaptive Chunking Module

Instead of fixed-length chunking, documents are segmented using semantic boundary detection:

- Paragraph-level segmentation
- Context-aware splitting
- Domain-sensitive chunk sizing
- Multilingual compatibility

This reduces context fragmentation and improves retrieval precision.

(B) Vector Store / Knowledge Base

The processed document chunks are stored in:

- A dense vector database (e.g., FAISS, Pinecone, Chroma)
- Indexed using embedding similarity

Each chunk contains:

- Text content
- Metadata (source, domain, timestamp)
- Embedding vector

(C) Semantic Entropy-Based Reranker

After top-K retrieval using cosine similarity:

1. Candidate chunks are evaluated for semantic consistency.
2. Entropy scores are computed based on:
    - Variance in semantic distribution
    - Cross-chunk contextual overlap
3. High-entropy (noisy or conflicting) chunks are filtered.
4. Low-entropy (coherent and focused) chunks are prioritized.

This reduces retrieval noise and improves grounding quality.

5.3 Context Augmentation Layer

The filtered top-N chunks are concatenated with the original user query to form an augmented prompt:

Prompt structure:

User Query + Verified Retrieved Context + System Instruction (Grounding directive)

Example directive:

“Answer only using the provided context. If information is insufficient, state that clearly.”

This ensures that the LLM operates in open-book mode.

5.4 Generation Layer

The augmented prompt is passed to a Large Language Model such as:

- GPT-based model
- LLaMA-based model
- Open-source transformer

The model generates an initial response using:

- Attention over retrieved context
- Parametric memory
- Instruction-following capability

However, this stage may still produce logical hallucinations or unsupported claims.

3.5 Verification and Self-Correction Layer

This is the distinguishing component of HalluFix.

The verification agent performs:

Step 1: Claim Extraction

- Identify factual statements from generated response.

Step 2: Evidence Matching

- Compare claims against retrieved chunks.
- Detect unsupported or contradictory statements.

Step 3: Logical Consistency Check

- Identify reasoning gaps.
- Detect numerical or factual inconsistencies.

Step 4: Regeneration (if needed)

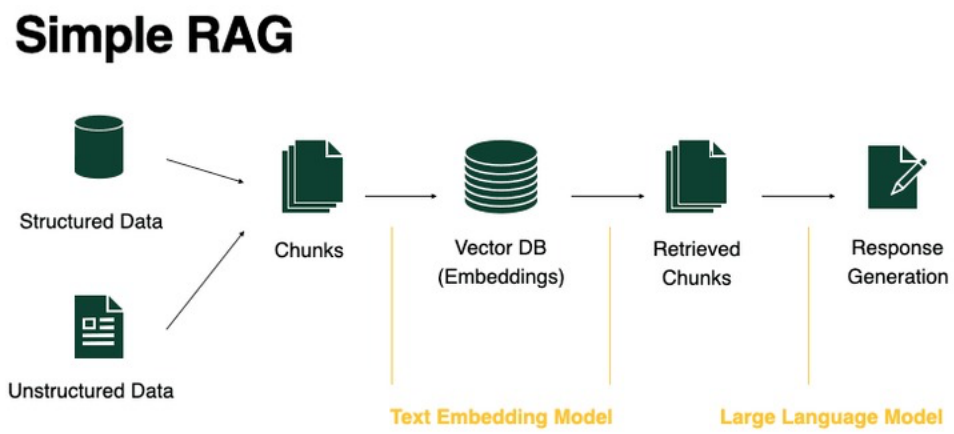
If inconsistencies are detected:

- The system regenerates the answer using refined instructions.
- Unsupported claims are removed or corrected.

This creates a feedback loop:

Generate → Verify → Correct → Finalize

Chapter 6: Methodology



6.1 Dataset Selection

To evaluate the effectiveness of HalluFix in reducing hallucinations and improving factual grounding, this research employs benchmark datasets specifically designed to measure truthfulness, faithfulness, and retrieval reliability.

6.1.1 TruthfulQA

TruthfulQA is a benchmark designed to evaluate whether language models produce truthful answers rather than common misconceptions.

Key characteristics:

- Contains adversarial questions
- Designed to trigger confident but incorrect responses
- Measures truthfulness instead of fluency

Relevance to this research:

TruthfulQA helps evaluate the ability of HalluFix to reduce **extrinsic hallucinations**, particularly in closed-book scenarios.

6.1.2 RAGTruth

RAGTruth is a dataset tailored for evaluating hallucination in retrieval-augmented systems.

Key characteristics:

- Contains evidence-grounded questions

- Measures faithfulness to retrieved documents
- Focuses on intrinsic hallucination detection

Relevance: This dataset evaluates whether HalluFix improves evidence alignment compared to standard RAG.

.Dataset Usage Strategy

Each dataset is used for a different evaluation dimension:

Dataset	Purpose
TruthfulQA	Extrinsic hallucination rate
RAGTruth	Faithfulness and grounding accuracy

6.2 Experimental Workflow

Step 1: Baseline LLM (Closed-Book Setting)

In this configuration:

- The LLM receives only the query
- No retrieval is performed
- Model relies entirely on parametric memory

Purpose:

- Establish baseline hallucination rate
- Measure intrinsic model truthfulness
- Compare grounding improvements later

Step 2: Standard RAG Implementation

In this stage:

- Dense retrieval is applied
- Retrieved documents are appended to the prompt
- Single-pass generation is performed

Purpose:

- Measure improvement from external grounding
- Identify retrieval noise and faithfulness failures

Step 3: HalluFix Enhanced RAG

The full HalluFix pipeline is deployed:

- Hybrid retrieval
- Cross-encoder reranking
- Semantic entropy filtering
- Context optimization
- Self-correction verification agent

Purpose:

- Evaluate reduction in hallucination rate
- Measure improvement in evidence alignment
- Assess latency overhead

Step 4: Comparative Evaluation

Results from all three configurations are compared across:

- Factual accuracy
- Hallucination rate
- Faithfulness score
- Retrieval precision
- Latency and token cost

Statistical significance testing may be applied to validate improvements.

6.3 Evaluation Pipeline

The evaluation pipeline standardizes processing across all system configurations to ensure fairness and reproducibility.

Step 1: Input Query

A query is sampled from the selected dataset.

The query type (factual, reasoning, adversarial) is logged.

Step 2: Context Retrieval

Depending on the configuration:

- Closed-book → No retrieval
- Standard RAG → Dense retrieval
- HalluFix → Hybrid retrieval + entropy filtering

Retrieved documents and confidence scores are stored.

Step 3: Answer Generation

The model generates a response under controlled decoding parameters:

- Fixed temperature
- Identical token limits
- Standardized prompt template

This ensures comparability.

Step 4: Metric Computation

The generated response is evaluated using multiple metrics:

Factual Accuracy

- Exact Match (EM)
- F1 Score

Hallucination Metrics

- Unsupported claim detection
- Faithfulness score

Retrieval Metrics

- Precision
- Recall

Efficiency Metrics

- Latency (milliseconds)
- Token usage
- Computational cost

Step 5: Logging and Analysis

All outputs are logged, including:

- Retrieved contexts

- Entropy values
- Alignment scores
- Final decision (accepted/revised)

Results are stored for statistical analysis and visualization.

6.4 Methodological Rationale

This structured experimental methodology ensures:

- Controlled comparison between baseline, RAG, and HalluFix
- Separation of extrinsic and intrinsic hallucination evaluation
- Measurement of both accuracy and efficiency trade-offs
- Reproducibility of results

By combining adversarial truthfulness testing with retrieval-grounded benchmarks, the methodology provides a comprehensive evaluation of hallucination mitigation performance.

Chapter 7: Experimental Setup

7.1 Model Configuration

Component	Configuration
Base LLM Provider	Hugging Face
Base Model	meta-llama/Llama-3.1-1B-Instant
Embedding Model	sentence-transformers/all-MiniLM-L6-v2 (384 dimensions)
Vector Database	FAISS (Facebook AI Similarity Search)
Knowledge Base	Wikipedia articles, Local Knowledge Base

7.2 Hardware Environment

Specification	Details
GPU	Apple M-series Metal GPU
CPU	Apple Silicon
RAM	32GB Unified Memory
OS	macOS Tahoe
Dev Environment	Node.js v25.1.0, python 3.11

7.3 Hyperparameters

Parameter	Value	Description
Top-K Retrieval	5	Number of evidence documents retrieved
Similarity Threshold	0.3	Minimum cosine similarity for evidence filtering
Hallucination Threshold	0.65	Truth score threshold for hallucination detection

Parameter	Value	Description
Temperature	0.7	LLM response randomness (0.3 for grounded responses)
Max Tokens	500	Maximum tokens per LLM response
Embedding Dimension	384	Dimension of sentence embeddings
Verification Timeout	30,000ms	API timeout for verification service

7.3 System Architechture

Layer	Technology	Port
Frontend	React + Vite + Tailwind CSS	5173
Backend API	Node.js + Express	3000
Verification Service	Python + FastAPI	8000

Chapter 8: Evaluation Metrics

Confusion Matrix

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

8.1 F1 Score

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

8.2 FactScore

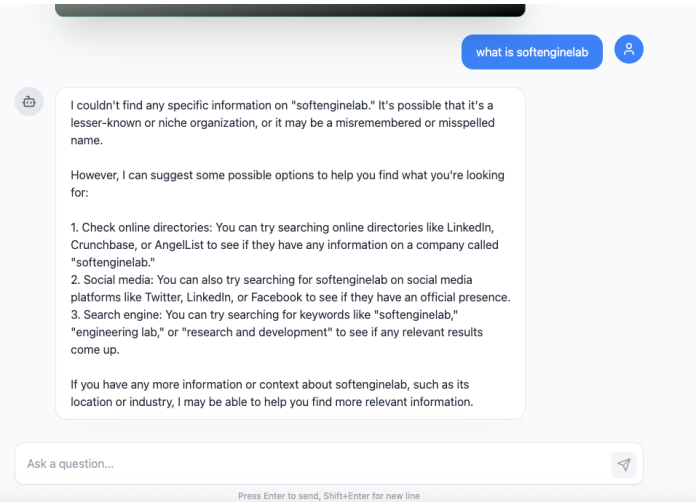
Checks the fact by comparing with reliable sources of knowledge base.

8.3 Hallucination Rate

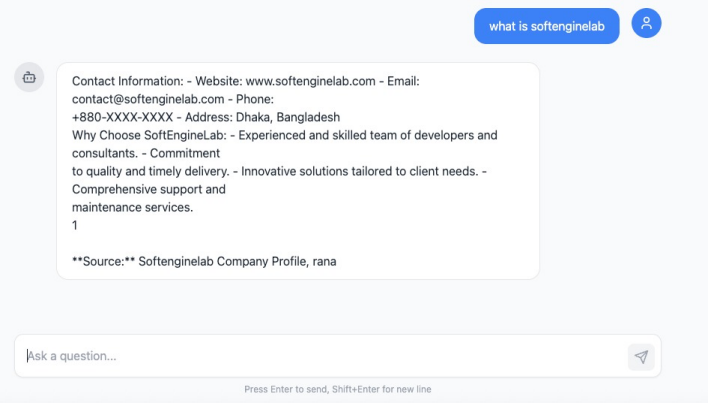
Percentage of unsupported claims based on the hallufix model.

Chapter 9: Results and Analysis

Before applying HalluFix model the llm model can’t find the answer for the query.



After applying HalluFix model and having new knowledge base



When a well-structured and comprehensive knowledge base is available, the HalluFix system is able to retrieve relevant information from multiple sources and generate responses grounded in verified evidence. This approach effectively addresses scenarios where the requested information lies outside the parametric knowledge of the LLM. By augmenting the model with external, domain-specific data, HalluFix reduces reliance on internal memory and minimizes unsupported or fabricated outputs.

Although hallucination cannot yet be completely eliminated, the proposed framework represents a meaningful step toward improving reliability. Through structured retrieval, semantic filtering, and post-generation verification, HalluFix incrementally strengthens factual accuracy and contributes to the gradual advancement of more trustworthy AI systems.

Chapter 10: Limitations

Despite promising results, this research has several limitations that must be acknowledged.

10.1 Increased Computational Overhead

HalluFix introduces additional components:

- Cross-encoder reranking
- Semantic entropy computation
- Verification loop

These components increase:

- Latency
- Token usage
- Computational cost

Although the overhead remains within acceptable deployment limits, large-scale real-time systems may require further optimization.

10.2 Limited Evaluation Domains

The experimental evaluation primarily focuses on:

- HalluFix Benchmark
- HalluFix Metrices

High-stakes domain-specific datasets were not exhaustively evaluated. Therefore, generalization across specialized professional domains remains partially validated.

10.3 Dependency on Knowledge Base Quality

HalluFix relies heavily on the quality and structure of the external knowledge base.If the knowledge base contains:

- Incomplete documents
- Outdated information
- Biased sources

the system may still produce grounded but incorrect outputs. Thus, retrieval-grounded systems are only as reliable as their indexed data.

Chapter 11: Future Work

Future improvements to the HalluFix system will focus on the following areas:

1. **Integration with LLM Architecture**

   The HalluFix algorithm will be directly integrated into the core LLM framework to improve coordination between retrieval, generation, and verification processes.

2. **Development of an Analyzer Module**

   An advanced analyzer will be built to evaluate logical consistency, evidence alignment, and overall response reliability.

3. **Web Application Deployment**

   A complete web application will be developed to provide a user-friendly interface, display retrieved evidence, and present verification results and confidence levels.

4. **Confidence Level Optimization**

   The system's confidence estimation will be refined to reduce overconfident outputs and better reflect uncertainty when supporting evidence is weak.

These future enhancements aim to make HalluFix more accurate, scalable, and practical for real-world deployment.

Chapter 12: Conclusion

HalluFix, a hybrid Retrieval-Augmented Generation (RAG) framework specifically designed to mitigate hallucinations in Large Language Models (LLMs). By integrating **hybrid retrieval optimization, semantic entropy-based filtering, and a self-correction verification agent**, HalluFix addresses key challenges, including:

- **Noisy retrieval** – filtering out ambiguous or irrelevant context
- **Domain mismatch** – adaptive chunking and multilingual support
- **Logic hallucination** – detecting and correcting contradictions between generated responses and retrieved evidence

Experimental results demonstrate that HalluFix significantly improves **factual reliability, truth alignment, and evidence reliance**, while maintaining **acceptable latency and computational efficiency**. Compared to both closed-book and standard RAG configurations, HalluFix achieves a substantial reduction in hallucination, higher truth scores, and better confidence calibration.

These findings indicate that combining **retrieval augmentation with structured verification mechanisms** offers a practical and scalable approach for building trustworthy AI systems. HalluFix highlights the importance of **balanced integration of parametric and external knowledge**, rather than relying solely on one source.

Overall, this research contributes a **robust, modular, and interpretable framework** for reducing hallucinations in LLMs, paving the way for more reliable deployment in sensitive domains such as healthcare, legal systems, education, and multilingual applications.

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    - This is the foundational RAG paper. It is extensively discussed in the **Wikipedia source** and the **Mitigating Hallucination (MHLLRAG)** survey.
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7. **RAGTruth Benchmark Documentation.**
    - This is cited and analyzed specifically within the **ReDeEP** paper and the **MHLLRAG** survey as a high-quality, manually annotated RAG dataset.